

Seen the right way, a page would let the agent solve up to 16 more tasks in 100 than the best fixed choice — yet nothing we trained beat always using the cheapest way on both success and cost.

THE AGENT LOOP, AND WHERE THE DECISION SITS

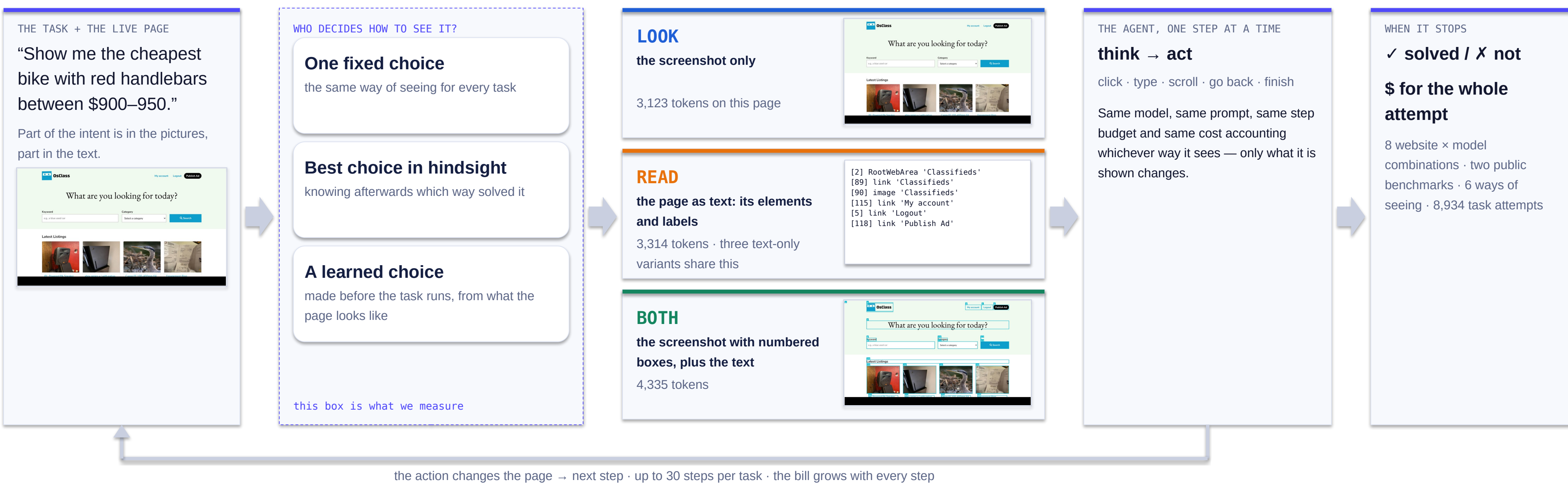
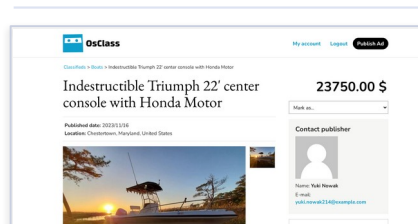
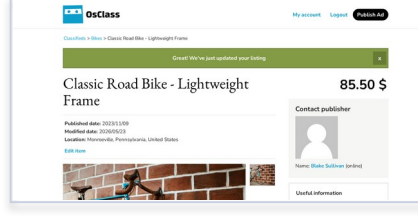
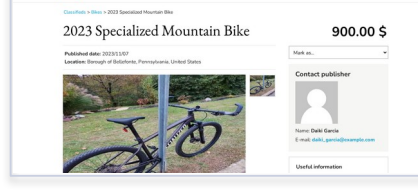


Fig 1. Everything is held fixed except how the page is shown to the agent. The dashed box is what this work measures — one fixed way of seeing, the best way in hindsight, or a way a model had to learn to choose — each judged on **both** success and cost. The laptop beside this poster replays the loop on the three tasks below.

WHAT THE SCREEN BESIDE YOU SHOWS

	LOOK	READ	BOTH
 “Navigate to the item on this page whose image is taken during a sunset.” ✓ 2 steps \$0.007	✗ 21 steps \$0.096	✓ 3 steps \$0.014	
 “Navigate to my listing of the blue bike and change the price to \$85.50 (including in the description).” ✗ 30 steps \$0.116	✓ 9 steps \$0.079	✗ 6 steps \$0.029	
 “Show me the cheapest bike with red handlebars between \$900-950.” ✗ 19 steps \$0.075	✗ 6 steps \$0.025	✓ 7 steps \$0.037	

One recorded attempt per way of seeing, B0 · classifieds. Every ✓ / ✗ came out the same on an independent rerun; steps and bill differ run to run — across all tasks a rerun flips 10–14% of outcomes.

WHY IT CANNOT BE LEARNED

A training example for “which way of seeing” exists only when the agent solves a task. Here the best single way solves just 2–36% of tasks, leaving typically 15–97 usable examples per setting — **the agents that would gain most from choosing produce the fewest examples to learn from.**

Shrinking the training data on purpose confirms scarcity is the mechanism, and prices it: the failing settings would need at least **2.1–4.2×** more tasks than the benchmarks contain.

Rerunning **one unchanged way of seeing** flips 10–14% of outcomes and by itself buys **2.0–7.6** more solved tasks in 100 (B0 · classifieds, six repeated ways, n=224); every gain on this sheet is read against that band, not against zero.

TAKEAWAY

Learning how to see is not only a modelling problem: whether it can be learned depends on how good the agent producing the examples already is. So, in this order: improve the agent, then collect reliable examples, then learn when to look.

Measured in the 2–36% success regime we observed. This need not hold for stronger agents.

RESULTS ACROSS 8,934 TASK ATTEMPTS

+16.35 in 100 IN HINDSIGHT, BEST OF 8 · VS ONE FIXED CHOICE
0 of 8 LEARNED CHOICES THAT BEAT ALWAYS-CHEAPEST
1 of 8 HINDSIGHT CHOICES THAT BEAT ALWAYS-CHEAPEST

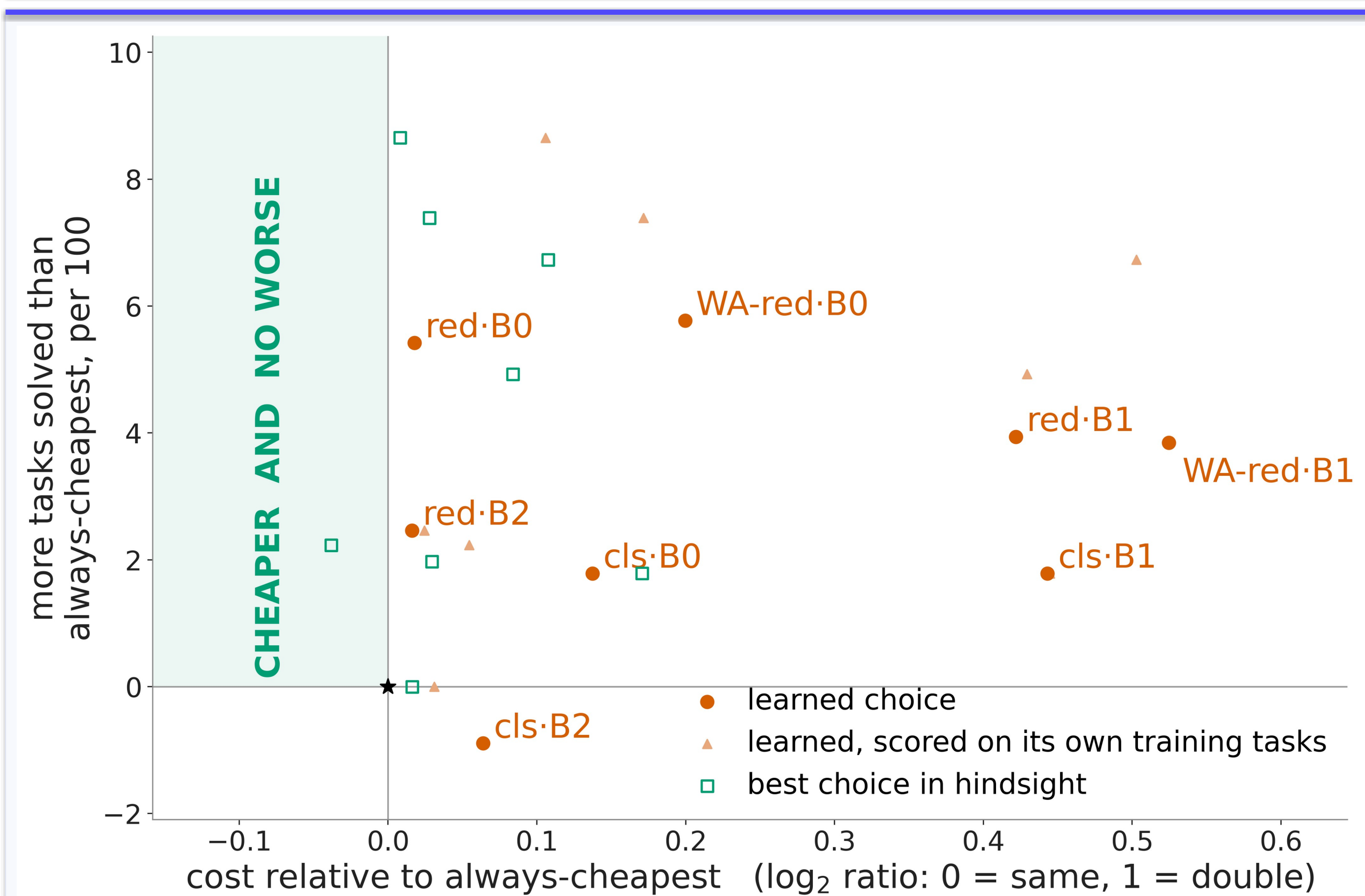


Fig 2. Every way of choosing, in every setting, compared with one fixed rule: **always use the cheapest way of seeing** (★). A win lands in the shaded region — cheaper *and* no worse. Always-cheapest is cheapest on average, not on every task, which is why a few points sit left of it. Learned choices are scored only on tasks they never saw.

Choosing well would pay. Picking the right way of seeing for each task after the fact would solve **3.45 to 16.35** more tasks in every 100 than the best single fixed choice, and spend 1.6–35.3% less — in all 8 settings.

Nothing we trained could do it. In **0 of 8** settings did a learned choice beat simply always using the cheapest way on both success and cost — and even choosing with hindsight manages that in only **1 of 8**.

What survives is a bound, not a method. Sending the tasks nobody solves to the cheapest way saves 9.5–30.6% at the same success in 8 of 8 — but that too needs hindsight, and plain always-cheapest usually saves more.

